Police station database

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"T4"

Our database is designed to be a representation of a police station where all the information about suspects, officers, evidence, cases, witnesses and victims are organized and managed to help track the criminals and to assign responsibilities to officers, the attributes come as follows:

- Officers: Tracks information about police officers including rank, specialization and contact information for the officer in charge of handling cases, obtaining evidence and investigating suspects.

- Cases: each case contains information like open/close dates, locations and type. Each case may have one or more suspects. Each case may also have one or more evidence.

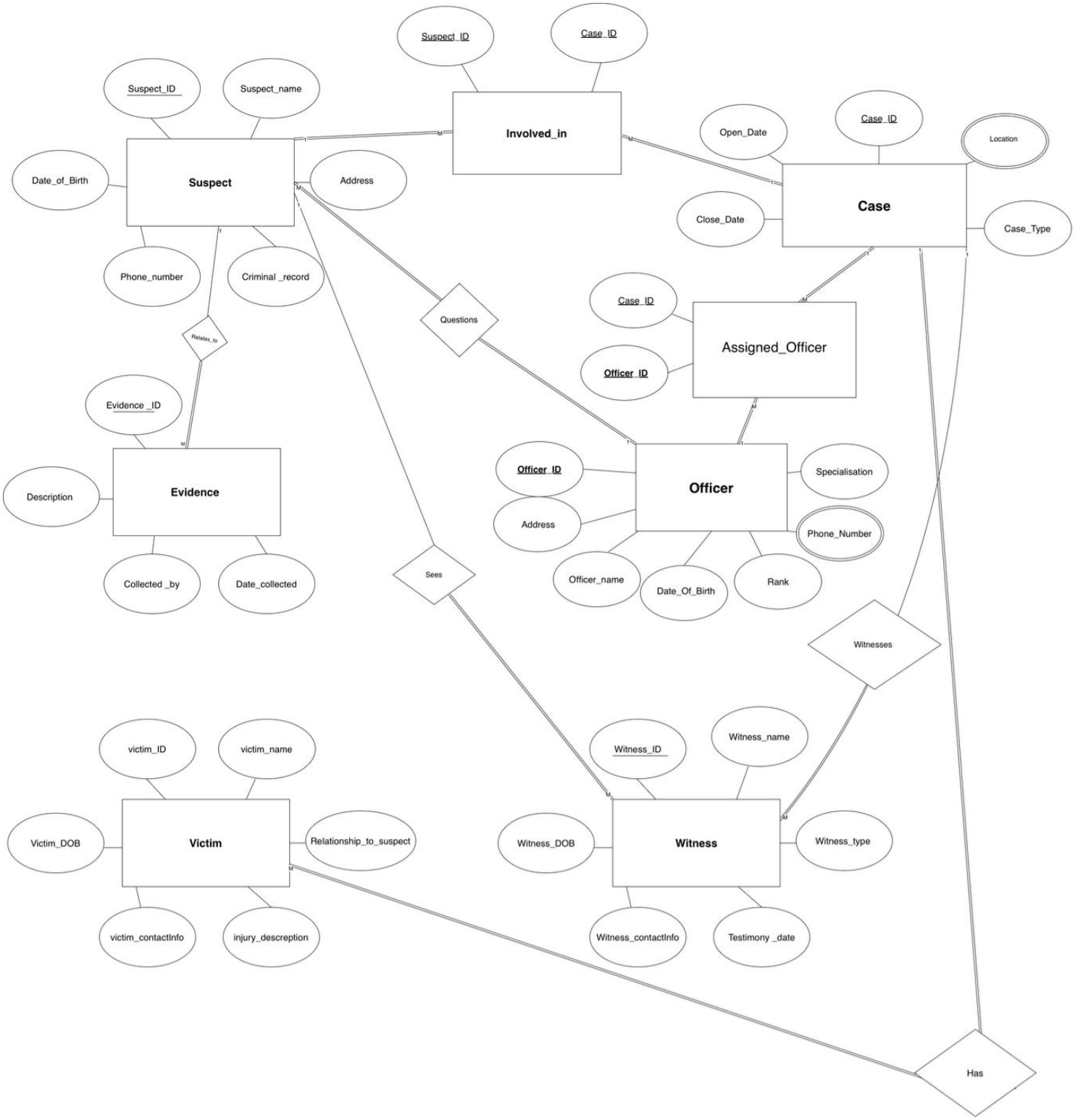
- Suspects: Stores information about people who are suspected of being involved in crimes, such as their name, date of birth, address, and phone number.

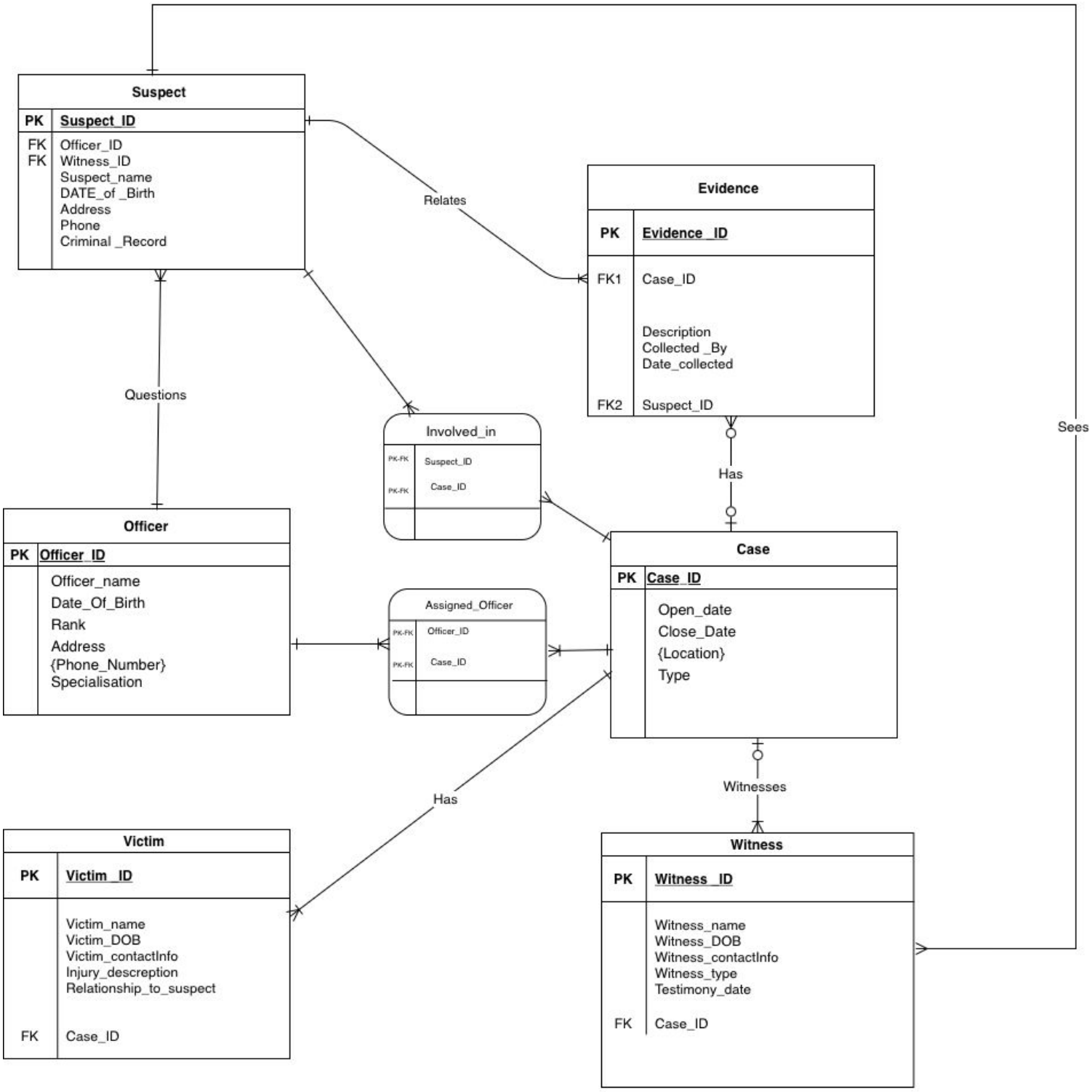
- Evidence: Contains details about evidence collected in cases, including description, collection date, the officer who collected it, and the related suspect.

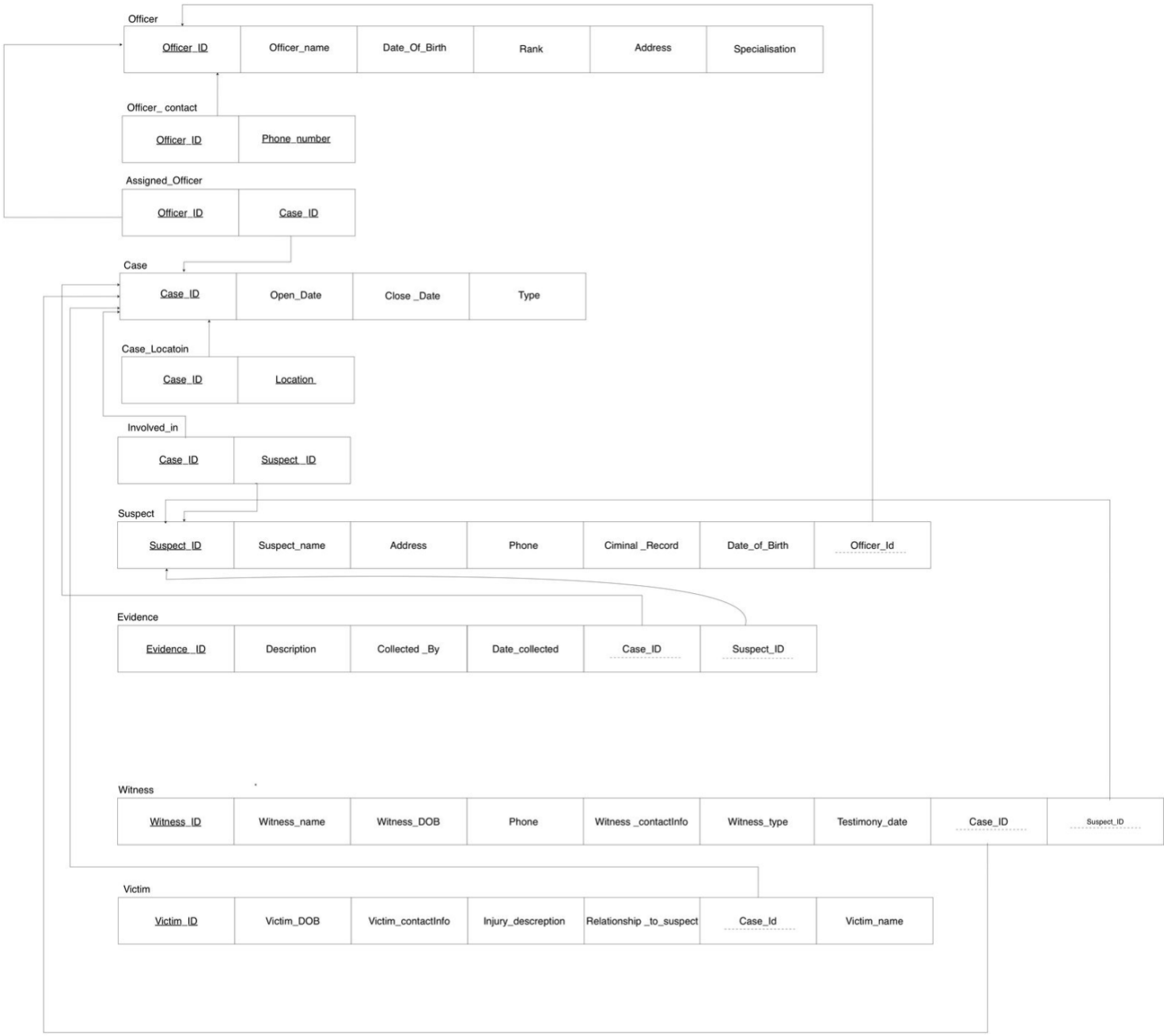
- Victim: stores information about individuals who have suffered harm or loss as a result of a criminal incident. Including the victim's unique identification number, personal details, contact information, the nature of any injuries sustained, and their relationship to the suspect involved in the case.

- Witness: Contains information about individuals who have observed or have relevant knowledge of a criminal incident. Including a unique identification number for each witness, personal and contact details, the type of witness, and the date on which their testimony was recorded.

Diagrams:


Suspect_ID
Case_ID
Suspect_ID
Suspect_name
Involved_in
Open_Date
Case_ID
Location
Date_of_Birth
Suspect
Address
Case
Close_Date
Case_Type
Phone_number
Criminal _record
Case_ID
Questions
Assigned_Officer
Relates_to
Officer_ID
Evidence _ID
Officer_ID
Specialisation
Description
Evidence
Officer
Address
Phone_Number
Collected _by
Date_collected
Sees
Officer_name
Date_Of_Birth
Rank
Witnesses
victim_ID
victim_name
Witness_ID
Witness_name
Victim_DOB
Victim
Relationship_to_suspect
Witness_DOB
Witness
Witness_type
victim_contactInfo
injury_descreption
Witness_contactInfo
Testimony _date
Has


Suspect
PK Suspect_ID
FK Officer_ID
FK Witness_ID
Suspect_name
DATE_of _Birth
Address
Phone
Criminal _Record
Relates
Evidence
PK Evidence _ID
FK1 Case_ID
Description
Collected _By
Date_collected
FK2 Suspect_ID
Questions
Involved_in
PK-FK Suspect_ID
PK-FK Case_ID
Has
Sees
Officer
PK Officer_ID
Officer_name
Date_Of_Birth
Rank
Address
{Phone_Number}
Specialisation
Assigned_Officer
PK-FK Officer_ID
PK-FK Case_ID
Case
PK Case_ID
Open_date
Close_Date
{Location}
Type
Has
Witnesses
Victim
PK Victim_ID
Victim_name
Victim_DOB
Victim_contactInfo
Injury_descreption
Relationship_to_suspect
FK Case_ID
Witness
PK Witness_ID
Witness_name
Witness_DOB
Witness_contactInfo
Witness_type
Testimony_date
FK Case_ID


Officer
Officer_ID
Officer_name
Date_Of_Birth
Rank
Address
Specialisation
Officer_ contact
Officer_ID
Phone_number
Assigned_Officer
Officer_ID
Case_ID
Case
Case_ID
Open_Date
Close _Date
Type
Case_Locatoin
Case_ID
Location
Involved_in
Case_ID
Suspect_ID
Suspect
Suspect_ID
Suspect_name
Address
Phone
Ciminal _Record
Date_of_Birth
Officer_Id
Evidence
Evidence_ID
Description
Collected _By
Date_collected
Case_ID
Suspect_ID
Witness
Witness_ID
Witness_name
Witness_DOB
Phone
Witness _contactInfo
Witness_type
Testimony_date
Case_ID
Suspect_ID
Victim
Victim_ID
Victim_DOB
Victim_contactInfo
Injury_descreption
Relationship _to_suspect
Case_Id
Victim_name

First Normal Form (1NF):

- All attributes are atomic (no multi-valued or composite attributes).

Second Normal Form (2NF):

- All non-key attributes are fully functionally dependent on the whole primary key.

Third Normal Form (3NF):

- No transitive dependency (non-key attributes do not depend on other non-key attributes).

SQL Statements:

/**/WITNESS TABLE/**/ 5

```
CREATE TABLE Witness (
Witness_ID NUMBER (10) not null,
Witness_name VARCHAR2 (100),
Witness_DOB DATE,
Phone NUMBER (10),
Witness_contactInfo VARCHAR2 (100),
Witness_type VARCHAR2 (50),
Testimony_date DATE,
Case_ID NUMBER (10),
Suspect_ID NUMBER (10),
CONSTRAINT fk_WCase_ID FOREIGN KEY (Case_ID) REFERENCES Case(Case_ID),
CONSTRAINT fk_Wsuspect_ID FOREIGN KEY (Suspect_ID) REFERENCES
SUSPECT(Suspect_ID),
CONSTRAINT pk_Witness_ID PRIMARY KEY (Witness_ID)
);
```

/**/VICTIM TABLE/**/ 4

```
CREATE TABLE Victim
(
Victim_ID NUMBER (10) not null,
Victim_name VARCHAR2 (100),
Victim_DOB DATE,
Injury_descreption VARCHAR2 (250),
Victim_contactInfo VARCHAR2 (100),
Relationship_to_suspect VARCHAR2 (50),
Case_ID NUMBER (10) not null,
CONSTRAINT fk_Case_ID FOREIGN KEY (Case_ID) REFERENCES Case(Case_ID),
CONSTRAINT pk_Victim_ID PRIMARY KEY (Victim_ID)
);
```

/**/EVIDENCE TABLE/**/ 10

```
CREATE TABLE Evidence (
Evidence_ID NUMBER(10) not null,
Description VARCHAR2(100),
Collected_By VARCHAR2(50),
Date_collected  DATE,
Case_ID NUMBER (10) not null,
Suspect_ID NUMBER(10) not null,
CONSTRAINT fk_Ecase_ID FOREIGN KEY(Case_ID) REFERENCES Case(Case_ID),
CONSTRAINT fk_Esuspect_ID FOREIGN KEY(Suspect_ID) REFERENCES Suspect(Suspect_ID),
CONSTRAINT pk_Evidence_ID PRIMARY KEY(Evidence_ID)
```

```
);
```

/**/SUSPECT TABLE/**/ 2

```
CREATE TABLE Suspect(
Suspect_ID NUMBER(10) not null,
Suspect_name VARCHAR2(100),
Address VARCHAR2(50),
Phone NUMBER (10),
Criminal_Record VARCHAR2(100),
Date_of_Birth DATE,
Officer_ID NUMBER (10),
CONSTRAINT fk_Officer_ID FOREIGN KEY(Officer_ID)REFERENCES Officer(Officer_ID),
CONSTRAINT pk_Suspect_ID PRIMARY KEY(Suspect_ID)
);
```

/**/INVOLVED_IN TABLE/**/ 9

```
CREATE TABLE Involved_in (
Case_ID NUMBER(10) not null,
Suspect_ID NUMBER(10) not null,
CONSTRAINT fk_Icase_Id FOREIGN KEY(Case_ID)REFERENCES Case(Case_ID),

CONSTRAINT fk_Isuspect_Id FOREIGN KEY(suspect_ID)REFERENCES suspect (suspect_ID),

CONSTRAINT pk_suspect_case_ID PRIMARY KEY(Suspect_ID,Case_ID)
);
```

/**/OFFICER TABLE/**/ 1

```
CREATE TABLE Officer (
   Officer_ID      NUMBER(10) NOT NULL,
   Officer_name    VARCHAR2(30),
   Date_Of_Birth   DATE,
   Rank            VARCHAR2(50),
   Address         VARCHAR2(50),
   Specialisation  VARCHAR2(40),
CONSTRAINT officerID_pk PRIMARY KEY (Officer_ID) );
```

/**/CASE TABLE/**/ 3

```
CREATE TABLE Case(
   Case_ID      NUMBER(10) NOT NULL,
   Type   VARCHAR2(30),
   Open_Date   DATE,
  Close_Date   DATE,
CONSTRAINT CaseID_pk PRIMARY KEY (Case_ID) );
```

/**/OFFICER_CONTACT TABLE/**/ 6

```
CREATE TABLE Officer_Contact (
   Officer_ID    NUMBER(10) NOT NULL,
   Phone_Number  NUMBER(10),
   CONSTRAINT OfficerContact_pk PRIMARY KEY (Officer_ID, Phone_Number),
   CONSTRAINT OfficerID_fk FOREIGN KEY (Officer_ID) REFERENCES Officer (Officer_ID)
);
```

/**/CASE_LOCATION TABLE/**/ 8

```
CREATE TABLE Case_location (
   Case_ID    NUMBER(10) NOT NULL,
   Location VARCHAR2(50),
CONSTRAINT CaseID_fk FOREIGN KEY (Case_ID) REFERENCES Case(Case_ID));
```

/**/ASSIGNED_OFFICER TABLE/**/ 7

```
CREATE TABLE Assigned_Officer(
   Officer_ID    NUMBER(10) NOT NULL,
   Case_ID  NUMBER(10) NOT NULL,
   CONSTRAINT OfficerAssigned_pk PRIMARY KEY (Officer_ID, Case_ID),
   CONSTRAINT fk_OfficerID FOREIGN KEY (Officer_ID) REFERENCES Officer (Officer_ID),
CONSTRAINT fk_CaseID FOREIGN KEY (Case_ID) REFERENCES Case (Case_ID)

);
```

/**/INSERTION PHASE/**/

```
insert into CASE values(9962946387,'murder',to_date('2019-8-1','YYYY-MM-DD'),to_date('2025-4-16','YYYY-MM-DD'));
insert into CASE values(9966253846,'kidnapping',to_date('2024-10-4','YYYY-MM-DD'),to_date('2024-12-2','YYYY-MM-DD'));
insert into CASE values(9961825390,'theft',to_date('2025-1-11','YYYY-MM-DD'),to_date('2025-1-15','YYYY-MM-DD'));
insert into CASE values(9962059374,'arson',to_date('2023-6-10','YYYY-MM-DD'),to_date('2023-7-19','YYYY-MM-DD'));
```

insert into CASE values(9662956483,'blackmail',to_date('2017-9-9','YYYY-MM-DD'),to_date('2022-2-8','YYYY-MM-DD'));

insert into OFFICER values(0197395668,'Thomas Blake',to_date('1978-3-8','YYYY-MM-DD'),'sergeant', '218 harbview-Boston', 'cold cases');

insert into OFFICER values(0192748597,'Jake Trivor',to_date('1980-7-22','YYYY-MM-DD'),'police officer', '103 elmrich-Chikago', 'civil affairs');

insert into OFFICER values(0192748500,'Brandin Kith',to_date('1974-11-2','YYYY-MM-DD'),'detictive', '390 malpwood-Seattle', 'murder cases');

insert into OFFICER values(0192749321,'Max Browin',to_date('1982-1-14','YYYY-MM-DD'),'lieutenant', '510 canehollow-Atlanta', 'cyber crimes');

insert into OFFICER values(0194048572,'Mason Harlow',to_date('1976-5-29','YYYY-MM-DD'), 'captin','742 ashbert-LosAngelos', 'fire investigations');

Insert into SUSPECT values(4509273640,'Jake Miller', '339 riverdalestreet-Chicago', 1029863988, 'armed robbery-shoplifting', to_date('2000-4-12','YYYY-MM-DD'), 192748597);

insert into SUSPECT values(4501298611,'Maria Court','209 sunsetavenue-Miami', 1928376603, 'hacking servers-data theft', to_date('2003-11-19','YYYY-MM-DD'), 192749321);

insert into SUSPECT values(4503382679,'Evan Lancaster','891 bronxboulevard-NYC', 1763974420, 'arson', to_date('1998-10-8','YYYY-MM-DD'), 194048572);

insert into SUSPECT values(4505927342,'Hazel Wood','662 windmerelane-LA', 1864732976, 'none', to_date('1990-1-3','YYYY-MM-DD'),197395668);

insert into SUSPECT values(4509265179,'Devon Harris','712 ironcladroad-Texas', 1552398607, 'assault-murder attempt', to_date('1985-7-4','YYYY-MM-DD'), 192748500);

insert into WITNESS values(4609283774,'Mia Davis', to_date('2001-7-31','YYYY-MM-DD'), 1883776492 ,'miaa99@gmail.com', 'eye witness', to_date('2025-1-11','YYYY-MM DD'), 9961825390,4509273640);
INSERT INTO WITNESS VALUES (0000000000, '', NULL, NULL, '', '', NULL, 9662956483,4501298611);

insert into WITNESS values(4602383029,'Ben Carter', to_date('2006-8-15','YYYY-MM-DD'), 1166253825 ,'benben738@gmail.com', 'police witness', to_date('2023-6-10','YYYY MM-DD'), 9962059374,4503382679);
insert into WITNESS values(4609929315,'Lucy Adams', to_date('2003-3-11','YYYY-MM-DD'), 1738466392 ,'lulu21@hotmail.com', 'informant ', to_date('2024-10-4','YYYY-MM DD'), 9966253846,4505927342);
insert into WITNESS values(4602376527,'Noah Parker', to_date('1999-4-7','YYYY-MM-DD'), 1749274005 ,'noahpa55@outlock.com', 'victim witness', to_date('2024-8-1','YYYY MM-DD'), 9962946387,4509265179);

insert into VICTIM values(0012389945,'emily Brown',to_date('2017-05-7','YYYY-MM-DD'),'bruises on wrists, face, signs of trauma','e.brown@gmail.com','neighbor',9966253846);

insert into VICTIM values(0012965318,'David Smith',to_date('2009-12-4','YYYY-MM-DD'),'emotional distress, anxiety, insomnia','david97@gmail.com','online friend',9662956483);

insert into VICTIM values(0012216906,'James Wilson',to_date('1995-1-10','YYYY-MM-DD'),'2nd-dgree burns , smoke inhalation','jajms@hotmail.com','Ex-partner',9962059374);

insert into VICTIM values(0012802582,'Anna White',to_date('2000-10-10','YYYY-MM-DD'),'Cut on head , scratches on arms','anna.white.64@outlock.com','stranger',9961825390);

insert into VICTIM values(0012321971,'Robert Clark',to_date('1980-5-7','YYYY-MM-DD'),'medical examiner report','robee812@ghotmail.com','Acquaintance',9962946387);

INSERT INTO EVIDENCE values(7774930276,'DNA of the victim','James Carl',to_date('2024-10-28','YYYY-MM-DD'),9966253846,4505927342);

INSERT INTO EVIDENCE values(7779538201,'Fingerprints','clara cooper',to_date('2025-1-11','YYYY-MM-DD'),9961825390,4509273640);

INSERT INTO EVIDENCE values(7773295318,'gasoline container in the suspects residence','mason harlow',to_date('2023-6-10','YYYY-MM-DD'),9962059374,4503382679);

INSERT INTO EVIDENCE values(7776930680,'Phone call recordings','Max browin',to_date('2022-1-29','YYYY-MM-DD'),9662956483,4501298611);

INSERT INTO EVIDENCE values(7778394211,'victims blood on the suspects clothing','leo miller',to_date('2019-8-21','YYYY-MM-DD'),9962946387,4509265179);

```
insert into ASSIGNED_OFFICER values(197395668, 9966253846);
insert into ASSIGNED_OFFICER values(192748597, 9961825390);
insert into ASSIGNED_OFFICER values(192748500, 9962946387);
insert into ASSIGNED_OFFICER values(192749321, 9662956483);
insert into ASSIGNED_OFFICER values(194048572, 9962059374);

insert into CASE_LOCATION values(9966253846, 'Fort Tryon Park');
insert into CASE_LOCATION values(9961825390, 'South Coast Plaza');
insert into CASE_LOCATION values(9962059374, 'Alamo Square');
insert into CASE_LOCATION values(9662956483, '209 sunsetavenue');
insert into CASE_LOCATION values(9962946387, 'Capitol Hill');

insert into INVOLVED_IN values(9962059374, 4503382679);
insert into INVOLVED_IN values(9662956483, 4501298611);
insert into INVOLVED_IN values(9966253846, 4505927342);
insert into INVOLVED_IN values(9961825390, 4509273640);
insert into INVOLVED_IN values(9962946387, 4509265179);

insert into OFFICER_CONTACT values(197395668, 1629373028);
insert into OFFICER_CONTACT values(192748597, 1420388469);
insert into OFFICER_CONTACT values(192748500, 1220386520);
insert into OFFICER_CONTACT values(192749321, 1600283682);
insert into OFFICER_CONTACT values(194048572, 1377020937);
```

/**/GET CASE BY OFFICER ID/**/

```
CREATE OR REPLACE PROCEDURE Get_Case_By_Officer_ID (
p_officer_ID IN VARCHAR2
)
AS
BEGIN
FOR v_cursrec IN (
SELECT case_ID
FROM Assigned_officer
WHERE OFFICER_ID = p_officer_ID
) LOOP
DBMS_OUTPUT.PUT_LINE('Case ID: ' || v_cursrec.case_ID);
END LOOP;
END;

EXEC Get_Case_By_Officer_ID(192749321);
```

/**/UPDATE OFFICER RANK/**/

```
CREATE OR REPLACE PROCEDURE Update_Officer_Rank (
p_officer_id IN NUMBER,
p_new_rank IN VARCHAR2
)
AS
BEGIN
UPDATE Officer
SET rank = p_new_rank
```

```
WHERE officer_id = p_officer_id;

COMMIT;
END;

EXEC Update_Officer_Rank(192749321, 'Officer');
```

//**//QUERIES//**//

```
SELECT Open_Date,case_id FROM CASE ORDER BY Open_Date ASC;

SELECT case_ID, COUNT(witness_name) AS number_of_witnesses
FROM WITNESS
GROUP BY CASE_ID;

SELECT
suspect.suspect_name,
officer.officer_name, officer.rank
FROM suspect
JOIN officer ON suspect.officer_id = officer.officer_id
WHERE officer.rank IN (
SELECT rank
FROM officer
WHERE rank = 'captin'
);

SELECT SUSPECT.suspect_name, SUSPECT.suspect_ID, WITNESS.witness_id
FROM SUSPECT
JOIN WITNESS ON SUSPECT.suspect_ID = WITNESS.SUSPECT_ID
WHERE WITNESS.WITNESS_ID = 0;
```

//**//TRIGGER//**//

```
CREATE OR REPLACE TRIGGER rank_update_msg
AFTER UPDATE ON officer
FOR EACH ROW
BEGIN
   DBMS_OUTPUT.PUT_LINE('Officer rank is updated to : ' || :NEW.rank);
END;
```

Screenshots of the tables after population:

CASE_ID	TYPE	OPEN_DATE	CLOSE_DATE
9966253846	kidnapping	04-OCT-24	02-DEC-24
9961825390	theft	11-JAN-25	15-JAN-25
9962059374	arson	10-JUN-23	19-JUL-23
9662956483	blackmail	09-SEP-17	08-FEB-22
9962946387	murder	01-AUG-19	16-APR-25
9968484848	murder	01-AUG-19	16-APR-25

CASE_ID	LOCATION
9966253846	Fort Tryon Park
9961825390	South Coast Plaza
9962059374	Alamo Square
9662956483	209 sunsetavenue
9962946387	Capitol Hill

OFFICER_ID	OFFICER_NAME	DATE_OF_BIRTH	RANK	ADDRESS	SPECIALISATION
197395668	Thomas Blake	08-MAR-78	sergeant	218 harbview-Boston	cold cases
192748597	Jake Trivor	22-JUL-80	police officer	103 elmrich-Chikago	civil affairs
192748500	Brandin Kith	02-NOV-74	detictive	390 malpwood-Seattle	murder cases
192749321	Max Browin	14-JAN-82	Lieutenant	510 canehollow-Atlanta	cyber crimes
194048572	Mason Harlow	29-MAY-76	captin	742 ashbert-LosAngelos	fire investigations

OFFICER_ID	PHONE_NUMBER
192748500	1220386520
192748597	1420388469
192749321	1600283682
194048572	1377020937
197395668	1629373028

OFFICER_ID	CASE_ID
192748500	9962946387
192748597	9961825390
192749321	9662956483
194048572	9962059374
197395668	9966253846

SUSPECT_ID	SUSPECT_NAME	ADDRESS	PHONE	CRIMINAL_RECORD	DATE_OF_BIRTH	OFFICER_ID
4509273640	Jake Miller	339 riverdalestreet-Chicago	1029863988	armed robbery-shoplifting	12-APR-00	192748597
4501298611	Maria Court	209 sunsetavenue-Miami	1928376603	hacking servers-data theft	19-NOV-03	192749321
4503382679	Evan Lancaster	891 bronxboulevard-NYC	1763974420	arson	08-OCT-98	194048572
4505927342	Hazel Wood	662 windmerelane-LA	1864732976	none	03-JAN-90	197395668
4509265179	Devon Harris	712 ironcladroad-Texas	1552398607	assault-murder attempt	04-JUL-85	192748500

CASE_ID	SUSPECT_ID
9662956483	4501298611
9962059374	4503382679
9966253846	4505927342
9962946387	4509265179
9961825390	4509273640

VICTIM_ID	VICTIM_NAME	VICTIM_DOB	INJURY_DESCREPTION	VICTIM_CONTACTINFO	RELATIONSHIP_TO_SUSPECT	CASE_ID
12389945	emily Brown	07-MAY-17	bruises on wrists, face, signs of trauma	e.brown@gmail.com	neighbor	9966253846
12965318	David Smith	04-DEC-09	emotional distress, anxiety, insomnia	david97@gmail.com	online friend	9662956483
12216906	James Wilson	10-JAN-95	2nd-dgree burns , smoke inhalation	jajms@hotmail.com	Ex-partner	9962059374
12802582	Anna White	10-OCT-00	Cut on head , scratches on arms	anna.white.64@outlock.com	stranger	9961825390
12321971	Robert Clark	07-MAY-80	medical examiner report	robee812@ghotmail.com	Acquaintance	9962946387

WITNESS_ID	WITNESS_NAME	WITNESS_DOB	PHONE	WITNESS_CONTACTINFO	WITNESS_TYPE	TESTIMONY_DATE	CASE_ID	SUSPECT_ID
4609283774	Mia Davis	31-JUL-01	1883776492	miaa99@gmail.com	eye witness	11-JAN-25	9961825390	4509273640
0	-	-	-	-	-	-	9662956483	4501298611
4602383029	Ben Carter	15-AUG-06	1166253825	benben738@gmail.com	police witness	10-JUN-23	9962059374	4503382679
4609929315	Lucy Adams	11-MAR-03	1738466392	lulu21@hotmail.com	informant	04-OCT-24	9966253846	4505927342
4602376527	Noah Parker	07-APR-99	1749274005	noahpa55@outlock.com	victim witness	01-AUG-24	9962946387	4509265179

EVIDENCE_ID	DESCRIPTION	COLLECTED_BY	DATE_COLLECTED	CASE_ID	SUSPECT_ID
7774930276	DNA of the victim	James Carl	28-OCT-24	9966253846	4505927342
7779538201	Fingerprints	clara cooper	11-JAN-25	9961825390	4509273640
7773295318	gasoline container in the suspects residence	mason harlow	10-JUN-23	9962059374	4503382679
7776930680	Phone call recordings	Max browin	29-JAN-22	9662956483	4501298611
7778394211	victims blood on the suspects clothing	leo miller	21-AUG-19	9962946387	4509265179

Screenshots of the queries:

OPEN_DATE	CASE_ID
09-SEP-17	9662956483
01-AUG-19	9962946387
01-AUG-19	9968484848
10-JUN-23	9962059374
04-OCT-24	9966253846
11-JAN-25	9961825390

CASE_ID	NUMBER_OF_WITNESSES
9966253846	1
9962946387	1
9961825390	1
9662956483	0
9962059374	1

SUSPECT_NAME	OFFICER_NAME	RANK
Evan Lancaster	Mason Harlow	captin

SUSPECT_NAME	SUSPECT_ID	WITNESS_ID
Maria Court	4501298611	0